GCSE · OCR · Computer Science

1 hour 27 questions

Defensive Design & Testing

Defensive Design & Testing

Defensive Design Considerations / Input Validation / Program Maintainability / Testing Programs / Identify Syntax & Logic Errors in Testing / Selecting & Using Suitable Test Data

Easy (8 questions)	/14
Medium (9 questions)	/35
Hard (10 questions)	/35
Total Marks	**/84**

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Easy Questions

1 Jack is writing a program to add up some numbers. His first attempt at the program is shown.

```
a = input("Enter a number")

b = input("Enter a number")

c = input("Enter a number")

d = input("Enter a number")

e = input("Enter a number")

f = (a + b + c + d + e)

print(f)
```

Give two ways that the maintainability of this program could be improved.

.....

..... **(2 marks)**

2 A program should only allow values from 0 to 300 inclusive as valid inputs. If the data entered breaks this validation rule, an error message is displayed.

Complete the following **test plan** for the program**.**

Test data	Test type	Expected result
25	Normal	Value accepted
	Invalid	Invalid input message displayed
300	Boundary	

(2 marks)

3 (a) The algorithm for one section of a vending machine program is shown in pseudocode.

```
if money >= price then
venditem()
giveChange(money – price)
else
print("Error – not enough money inserted")
endif
```

When writing the program **maintainability** was considered.

Identify two ways that the program has been made more maintainable.

.....

..... **(2 marks)**

(b) **Give one additional** way that the **maintainability** of the program can be improved.

..... **(1 mark)**

4 A program allows users to search for and watch videos.

Users give a rating to the videos they watch.

Describe **one** method of defensive design that can be used when creating the program.

.....

..... **(2 marks)**

5 **State** one reason why a programmer should anticipate misuse when designing a program

..... **(1 mark)**

6 **Define** the term *input validation*

..... **(1 mark)**

7 **State** two reasons why programs are tested before they are released

.....

..... **(2 marks)**

8 **Identify** the type of testing that is carried out repeatedly during program development

..... **(1 mark)**

Medium Questions

1 Customers at a hotel can stay between 1 and 5 (inclusive) nights and can choose between a basic room or a premium room.

Complete the following test plan to check whether the **number of nights** is validated correctly.

Test data (number of nights)	Type of test	Expected output
2		ALLOWED
	Boundary	ALLOWED
	Erroneous / Invalid	NOT ALLOWED

(3 marks)

2 The area of a circle is calculated using the formula π × r2 where π is equal to 3.142 and r is the radius.

A program is written to allow a user to enter the radius of a circle as a whole number between 1 and 30, then calculate and output the area of the circle.

```
01 radius = 0

02 area = 0.0

03 radius = input("Enter radius")

04 if radius < 1 OR radius > 30 then

05 print("Sorry, that radius is invalid")
```

```
else
area = 3.142 * (radius ^ 2)
print (area)
endif
```

Explain, **using examples** from the program, **two ways** to improve the **maintainability** of the program.

1	
2	

.....

.....

.....

..... **(4 marks)**

3 The program should only allow values from 0 to 300 inclusive as valid inputs. If the data entered breaks this validation rule, an error message is displayed.

Complete the following program to output "Invalid input" if the data does not meet the validation rule.

You must use either:

- OCR Exam Reference Language, or
- a high-level programming language that you have studied.

```
mins = input("Enter minutes played: ")

if mins < 0 ........................ mins ........................ then

.............................. ("Invalid input")

endif
```

.....

.....

..... **(3 marks)**

4 A teacher writes an algorithm to store the name of the game a student plays each night (for example "OCR Zoo Simulator").

variable.length returns the number of characters in variable. variable.upper returns the characters in variable in upper case.

```
valid = false

while(valid == false)

gameName = input("Enter the game name")

if (gameName.length > 0) AND (gameName.length < 20)

gamesPlayed = gameName.upper

valid = true
```

```
print("Valid game name")

else

print("Game name is not valid")

endif

endwhile
```

The algorithm needs **testing** to make sure the **IF-ELSE statement** works correctly.

Identify two different pieces of test data that can be used to test different outputs of the algorithm.

Give the output from the program for each piece of test data.

Test data	Output

(4 marks)

5 (a) OCRBlocks is a game played on a 5 × 5 grid. Players take it in turns to place blocks on the board.

The board is stored as a **two-dimensional (2D) array** with the **identifier** gamegrid

Fig. 6.1 shows that players A and B have placed three blocks each so far.

	0	1	2	3	4
0	A			B	
1					
2		B			
3	A		B		
4			A		

The **function** checkblock() checks whether a square on the board has been filled. When checkblock(4,2) is called, the value "A" is returned.

```
function checkblock(r,c)

if gamegrid[r,c] == "A" or gamegrid[r,c] == "B" then

outcome = gamegrid[r,c]

else

outcome = "FREE"

endif

return outcome

endfunction
```

When checkblock(-1,6) is called, an error is produced

State why this **function** call will **produce an error**.

..... **(1 mark)**

(b) **Describe how validation** could be added in to the checkblock() function **to stop this error** from occurring.

.....

.....

..... **(3 marks)**

6 (a) A vending machine has the following options available.

Item code	Item name	Price
A1	Crisps, bacon flavour	£0.75
A2	Crisps, salted	£0.75
B1	Chocolate bar	£0.90
C1	Apple pieces	£0.50
C2	Raisins	£0.85

Users insert coins into the vending machine and then enter the two character item code of their selection. If the user has inserted enough money, the vending machine will release the chosen item and output any change required. If the user enters an invalid item code then a suitable error message is displayed.

The vending machine is **tested** before it is released.

Explain the purpose of testing the vending machine.

.....

..... **(2 marks)**

(b) **Describe the difference** between **iterative** testing and **final** testing.

.....

..... **(2 marks)**

(c) **Complete** the following **test plan** for the vending machine

Code entered	Money inserted	Expected result
B1	£1	Chocolate bar served, £0.10 change given
	£0.85	Raisins served, no change given
C1		Error – not enough money inserted
C3	£0.75	

(3 marks)

7 (a) Louise writes a program to work out if a number entered by the user is odd or even. Her first attempt at this program is shown.

```
01 num = input("enter a number")

02 if num MOD 2 >= 0 then

03     print("even")

04 else

05     pritn("odd")

06 endif
```

The program contains a **syntax error** on line **05**.

State what is meant by a syntax error.

..... **(1 mark)**

(b) **Give a corrected** version of line **05** that fixes the syntax error.

..... **(1 mark)**

8 Elliott plays football for OCR FC. He wants to create a program to store the results of each football match they play and the names of the goal scorers. Elliott wants individual players from the team to be able to submit this information.

Describe two examples of **defensive design** that should be considered when developing this program.

1	
2	

.....

.....

.....

..... **(4 marks)**

9 An algorithm stores the position of a character on a straight line as an integer.

A user can move the character left or right.

The following algorithm:

- Generates one random number between 1 and 512 (inclusive) and stores it as position.
- Prompts the user to input a direction to move ("left" or "right").
- Continues to take input until a valid direction is entered.

```
p ← RANDOM_INTEGER(1, 512)
OUTPUT "The position is ", p
a ← ""
WHILE a ≠ "left" AND a ≠ "right"
   a ← USERINPUT("Enter direction, left or right:")
ENDWHILE
```

Describe **two** ways to improve the maintainability of the algorithm.

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.....

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..... **(4 marks)**

Hard Questions

1 A hotel has nine rooms that are numbered from room 0 to room 8.

The number of people currently staying in each room is stored in an **array** with the **identifier** room.

The index of room represents the room number.

Array room

Index	0	1	2	3	4	5	6	7	8
Data	2	1	3	2	1	0	0	4	1

The following program **counts** how many people are currently staying in the hotel.

```
for count = 1 to 8

total = 0

total = total + room[count]

next count

print(total)
```

When **tested**, the program is found to contain **two logic errors**.

Describe how the program **can be refined** to remove these logic errors.

.....

..... **(2 marks)**

2 (a) Louise writes a program to work out if a number entered by the user is odd or even. Her first attempt at this program is shown.

```
01 num = input("enter a number")

02 if num MOD 2 >= 0 then

03 print("even")

04 else

05 pritn("odd")

06 endif
```

The program contains a **logic error** on line **02**.

State what is meant by a **logic error**.

..... **(1 mark)**

(b) **Give a corrected version** of line **02** that **fixes** the logic error.

..... **(1 mark)**

3 **Describe the purpose** of **input validation** in defensive programming.

.....

..... **(2 marks)**

4 **Write** a Python **function** that validates if an input string **is a valid email address** (contains "@" and ".").

.....

.....

.....

.....

..... **(5 marks)**

5 **Describe** the **differences** between authentication and input validation.

.....

.....

.....

..... **(4 marks)**

6 (a) The height a student jumps in the high jump needs to be input and validated.

The height is entered in centimetres (cm) and must be between 40.0 and 180.0 inclusive.

Write an algorithm to:

- take the height jumped as input
- output **"VALID"** or **"NOT VALID"** depending on the height input

You must use either:
• **OCR Exam Reference Language**, or
• a **high-level programming language** that you have studied.

.....

.....

.....

..... **(4 marks)**

(b) The algorithm is tested using a range of tests.

Complete the table to identify an example of test data for each type of test.

Test data (height jumped in cm)	Type of test	Expected output
	Normal	"VALID"
	Boundary	"VALID"
	Erroneous	"NOT VALID"

(3 marks)

7 **Evaluate** how the use of sub-programs (functions or procedures) can improve the maintainability of a large program

.....

.....

..... **(3 marks)**

8 **Justify** why testing for unauthorised access is an important part of a test plan for a banking app

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..... **(3 marks)**

9 **Compare** iterative testing and final testing in terms of when they are carried out and what they aim to achieve

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.....

..... **(4 marks)**

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10 A program accepts values between 1 and 10

Explain why the values **1** and **11** should both be included in a test plan

.....

.....

..... **(3 marks)**